Fiscal Year:	FY 2007	**Task Last Updated:**	FY 06/06/2007
PI Name:	Hall, Eric J Ph.D., D.Sc.		
Project Title:	Individual Genetic Susceptibility		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	New York	**State:**	NY
Zip Code:	10032	**Congressional District:**	15
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	02-OBPR-02
Start Date:	05/19/2003	**End Date:**	06/01/2008
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 6/1/2008 per P. Denkins (6/07) NOTE: end date changed to 5/31/2007 per J. Dardano (5/06)		
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):	Brenner, David J (Columbia University) Smilenov, Lubomir (Columbia University)		
Grant/Contract No.:	NAG 9-1519		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The hypothesis upon which this proposal is based is that the human population is not homogeneous in radiosensitivity, but that radiosensitive sub-groups exist which would suffer an increased incidence of detrimental radiation effects, and distort the shape of the dose response relationship. The radiosensitivity of these groups depends on the expression levels of specific proteins. In a previous proposal funded by NASA we used knock-out mice to show unequivocally that haploinsufficiency for the ATM gene resulted in increased sensitivity to gamma-rays for both ocular cataracts (a deterministic effect) and oncogenic transformation in embryo fibroblasts (a stochastic effect). It would be extraordinarily difficult and expensive to conduct similar experiments with the high energy Fe-59 ions of principal interest to NASA, because of the need to breed a large number of animals, all of the appropriate age, to be available on the few occasions each year when Fe-59 ions are available at the Brookhaven Laboratory. It would be even more difficult and expensive to expand the study to include a number of other genes that are suspected of being involved with radiosensitivity, and to investigate the possible interaction of these various genes. We proposed instead to screen cells heterozygous for a number of genes; Atm, BRCA1, BRCA2 and rad9, as well as double heterozygotes for pairs of these genes. Wild type animals as well as heterozygotes and double heterozygotes will be exposed to graded doses of ?-rays at Columbia or Fe-56 heavy ions at Brookhaven National Laboratory, and thymocytes assayed for early apoptotic death. By this means we plan to discover which genes or combinations of genes, are important in conferring radiosensitivity. For genes, or combinations of genes that look promising for this apoptotic endpoint, more extensive studies would be carried out with the more relevant endpoint oncogenic transformation using the embryo fibroblasts. This strategy is based on preliminary studies with gamma-rays and cell lines with a mutated gene (namely Atm) which show that a reduction in radiation-induced apoptosis is mirrored by an increase in oncogenic transformation.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The dose limits proposed for astronauts in space flight are different from those on the ground, though the basic philosophy of limiting the life-time risk of stochastic effects to 3% is preserved. There are several reasons for the differences, including the limited number of individuals exposed, the practicality of shielding and the presence of more urgent risks. Nevertheless the purpose of radiation protection remains to prevent deterministic effects of clinical significance and limit stochastic effects to acceptable levels. The levels proposed are based on the assumption that the human population is uniform in its radiosensitivity. Using two endpoints, namely cataracts as an important and relevant deterministic effect and oncogenic transformation in cultured embyro cells as a surrogate for carcinogenesis, the most relevant stochastic effect, we have shown that mice heterozygous for the atm gene are more radiosensitive than wild type animals. 1 to 3% of the US population are AT heterozygotes, and it would not be prudent to subject such individuals to substantial doses of radiation.
Task Progress:	The mice used in this project include (a) animals carrying a mutation in one allele of the Atm gene. These animals came from Dr. Philip Leder at Harvard. (b) Animals carrying a mutation in the BRCA1 gene, which leads to a truncated protein; this is similar to the mutation commonly found in the human case. These animals were produced by Dr. Thomas Ludwig at the Columbia University Herbert Irving Comprehensive Cancer Center. (Ludwig et al. a, b). In addition, we have cross-bred mice heterozygous for the above genes to produce double-heterozygotes. An example is Atm het/BRCA1 het. Pregnant females (of each genetic background, ie wild type, AT heterozygote, BRCA1 heterozygote or double heterozygote) were sacrificed on day 14 of gestation. The embryos were surgically removed, minced, trypsinized and plated in 10 cm plates. 4 days later, the cells were genotyped and used for transformation assays as follows. Exponentially growing mouse embryo fibroblasts (MEF) received a dose of 0.5 Gy of 56Fe – 1GeV/n in an acute exposure, and controls were sham-irradiated. MEF were then plated in 10 cm plates at a density of 6,000 cells/plate over a feeder layer of 70,000 cells prepared from the same embryo but irradiated previously with a supralethal dose of 30Gy. After 2 weeks of growth in DMEM medium supplemented with 10% fetal bovine serum at 37° C in a 5% CO2 air-humidified incubator, cells were fixed, stained, and yields of transformed clones scored. The scoring criteria were developed and examined by preliminary experiments, where embryo cells were irradiated and plated with the same density. The clones which seemed dense and had stellate-shaped piled cells were photographed and isolated with cloning cylinders. These clones were expanded and injected into nude mice. Those that caused the development of tumors were designated as transformed. Clones that matched their shape and dimensions were scored as transformed in later experiments. Plating efficiency, cell surviving fractions, and the spontaneous and radiation-induced frequency of transformation were determined. Atm heterozygotes showed twice the transformation frequency of the wild type cells following a dose of 0.5 By of heavy ions. Brca1 heterozygotes showed a similar transformation frequency as the Atm hz, however, the interesting point to note is that the double heterozygote Atm/Brca1, show no increase over Brca1hz or Atm hz alone. This is in marked contrast to the case for the double heterozygote Atm/rad 9, where the transformation frequency was much larger than for either single het.
Bibliography Type:	Description: (Last Updated: 10/26/2023)
Articles in Peer-reviewed Journals	Hall EJ. "Intensity-modulated radiation therapy, protons, and the risk of second cancers." Int J Radiat Oncol Biol Phys. 2006 May 1;65(1):1-7. Review. PMID: 16618572 , May-2006
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Articles in Peer-reviewed Journals	Travis LB, Rabkin CS, Brown LM, Allan JM, Alter BP, Ambrosone CB, Begg CB, Caporaso N, Chanock S, DeMichele A, Figg WD, Gospodarowicz MK, Hall EJ, Hisada M, Inskip P, Kleinerman R, Little JB, Malkin D, Ng AK, Offit K, Pui CH, Robison LL, Rothman N, Shields PG, Strong L, Taniguchi T, Tucker MA, Greene MH. "Cancer survivorship—genetic susceptibility and second primary cancers: research strategies and recommendations. " J Natl Cancer Inst. 2006 Jan 4;98(1):15-25. Review. PMID: 16391368 , Jan-2006